

L4a: Trading Rules Using Binomial Lattice Models

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you will be able to:

- **Evaluate a stock trade using NPV**: derive the scaled NPV and explain how sale price, holding period, and benchmark affect it.
- **Compute terminal target probabilities**: find the minimum up-move count needed to exceed a target and add the binomial probabilities.
- **Extend the lattice to m outcomes per step**: use branch counts to calculate node prices, probabilities, and state counts.

Lectures and Examples

Lecture: Trading Rules Using Binomial Lattice Models

Example: Binomial Lattice Terminal-Target Probabilities

What is the probability of exceeding a target scaled NPV? We estimate the lattice parameters from historical data and compare probabilities across targets.

Example: N-Ary Recombining Lattice Models

What changes when we allow more price movements per step? We estimate branch factors and probabilities, build the lattice, and examine its terminal outcomes.

From Bid and Ask Quotes to a Lattice Price

Citadel Securities is a **market maker**: it stands ready to buy from sellers and sell to buyers.

At time t , its quotes are:

- **Bid** b_t : the price at which it will buy.
- **Ask** a_t : the price at which it will sell.

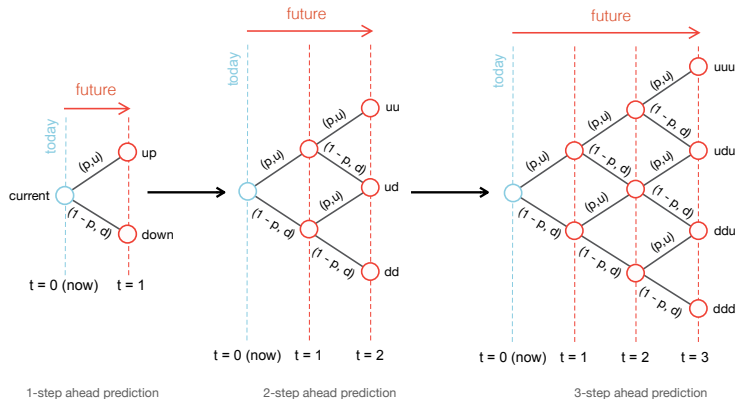
The spread s_t and midpoint m_t are:

$$\underbrace{s_t = a_t - b_t}_{\text{bid-ask spread}}, \quad \underbrace{m_t = \frac{a_t + b_t}{2}}_{\text{midpoint}}.$$

For today's model, we replace the bid and ask with one frictionless price S_t . The lattice assigns probabilities to its possible future values.

A Recombining Price Lattice

Each step lasts Δt years. Up and down branches multiply the price by u and d ; paths with the same counts reach the same node.



The Binomial Lattice Model

Start at price $S_0 > 0$. Each step multiplies the price by u with real-world probability p , or by d with probability $1 - p$, where $u > d > 0$ and $0 < p < 1$.

Assume **independent moves** and fixed (u, d, p) . Let t count steps and K_t count up moves. Then $K_t \sim \text{Binomial}(t, p)$:

$$\underbrace{S_{t,k} = S_0 u^k d^{t-k}}_{\text{node price}}, \quad \underbrace{\mathbb{P}(K_t = k) = \binom{t}{k} p^k (1-p)^{t-k}}_{\text{node probability}}, \quad k = 0, \dots, t.$$

Each level contains $t + 1$ price states. Independent moves with fixed parameters do not capture volatility clustering.

The Trade We Will Evaluate

Suppose we buy shares now and sell them after a chosen holding period.

- Buy $n_0 > 0$ shares at price $S_0 > 0$ USD/share at time zero.
- Hold for N steps of duration $\Delta t > 0$ years; sell at $T = N\Delta t$ for S_T USD/share.
- Discount the sale proceeds using a constant annual benchmark growth rate g_y , measured in inverse years.

We assume no dividends, transaction fees, or bid–ask spread.

Net Present Value of a Long Position

Use the continuously compounded benchmark g_y for yield y : a risk-free rate or an assumed growth rate for another investment. The trade's NPV is:

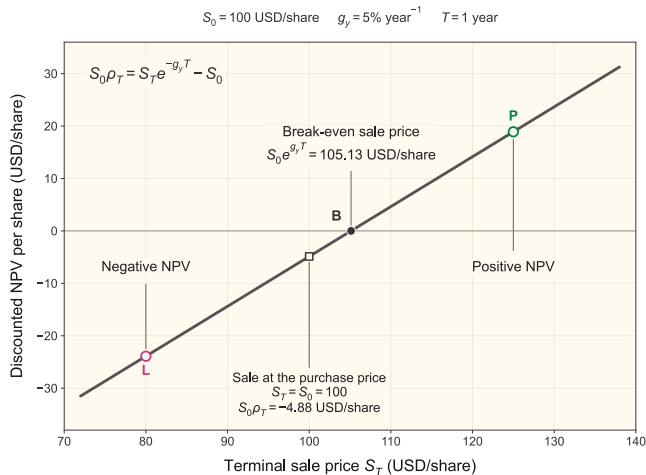
$$\underbrace{\text{NPV}(g_y, T)}_{\text{time-0 dollars}} = \underbrace{-n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted sale proceeds}}.$$

Divide by the initial investment to obtain the dimensionless scaled NPV:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\text{NPV}(g_y, T)}{n_0 S_0} = e^{-g_y T} \frac{S_T}{S_0} - 1.$$

If $\rho_T > 0$, the discounted sale proceeds exceed the purchase cost. Scaling removes the dependence on the number of shares.

Sale Price and Discounted NPV



Short Holding Periods

When $|g_y|T \ll 1$, the discount factor $e^{-g_y T}$ is close to one. The scaled NPV is approximately the fractional change in share price:

$$\underbrace{\rho_T}_{\text{scaled NPV}} \approx \underbrace{\frac{S_T}{S_0} - 1}_{\text{price return}}.$$

Buying at 100 USD/share and selling at 105 USD/share gives:

$$\frac{105}{100} - 1 = 0.05 = 5\%.$$

When discounting has little effect, the scaled NPV is approximately 5%.

Longer Holding Periods

As the holding period grows, discounting can have a substantial effect. The sale price that makes the scaled NPV zero satisfies:

$$e^{-g_y T} \frac{S_T}{S_0} - 1 = 0 \quad \Rightarrow \quad \underbrace{S_T = S_0 e^{g_y T}}_{\text{break-even sale price}} .$$

For $g_y > 0$, a positive NPV requires the sale price to exceed the purchase price grown at the benchmark rate.

We use the exact discounted expression for all holding periods. Each lattice price gives a scaled NPV with the same probability.

Terminal Target Probability

At the scheduled sale time $T = N\Delta t$, we calculate the probability that the scaled NPV exceeds a chosen target $\rho_\star$:

$$\mathbb{P}(\rho_T > \rho_\star).$$

A target $\rho_\star = 0.05$ means that the discounted sale proceeds must exceed the initial investment by more than 5%. Equality does not count.

We identify terminal nodes whose scaled NPV exceeds the target and add their probabilities. Each terminal price is determined by the number of up moves.

From Up Moves to Scaled NPV

The up-move count is $K_N \sim \text{Binomial}(N, p)$. A terminal node with k up moves has:

$$S_T = S_0 u^k d^{N-k}, \quad k = 0, \dots, N.$$

Substituting into the scaled NPV gives:

$$\rho(k) = u^k d^{N-k} e^{-g_y N \Delta t} - 1, \quad \rho_T = \rho(K_N).$$

Increasing k replaces a down factor d with an up factor u . Since $u/d > 1$, both the price and scaled NPV increase.

Once a node's scaled NPV exceeds the target, every larger up-move count succeeds.

Finding the Up-Move Threshold

For $\rho_\star > -1$, start with the strict target at a node with k up moves:

$$e^{-g_y N \Delta t} u^k d^{N-k} - 1 > \rho_\star.$$

Add one, remove the discount factor, and take natural logarithms:

$$k \ln(u/d) > \ln(1 + \rho_\star) + g_y N \Delta t - N \ln d.$$

Since $\ln(u/d) > 0$, the condition becomes $k > \tau(\rho_\star)$, where:

$$\underbrace{\tau(\rho_\star)}_{\text{up-move threshold}} = \frac{\ln(1 + \rho_\star) + g_y N \Delta t - N \ln d}{\ln(u/d)}.$$

Strict Targets Need the Next Integer

We need $K_N > \tau(\rho_\star)$. Since the up-move count is an integer, the smallest integer strictly above the threshold is:

$$\kappa = \lfloor \tau(\rho_\star) \rfloor + 1.$$

Here, $\lfloor x \rfloor$ is the greatest integer less than or equal to x .

- If $\tau = 2.4$, we need at least three up moves.
- If $\tau = 2$ exactly, we still need three. Two would only reach the target.

Example: A Three-Month Trade

Hold for $N = 3$ monthly steps: $\Delta t = 1/12$ year and $T = 1/4$ year. Use $u = 1.10$, $d = 0.90$, $p = 0.6$, and $g_y = 0.04 \text{ year}^{-1}$.

To exceed a target scaled NPV of 5%, the up-move threshold is:

$$\tau(0.05) = \frac{\ln(1.05) + 0.04(3/12) - 3 \ln(0.90)}{\ln(1.10/0.90)} \approx 1.868.$$

We need at least $\kappa = 2$ up moves. Check the two neighboring returns:

$$\rho(1) \approx -0.1179, \quad \rho(2) \approx 0.0782.$$

Two up moves exceed the 5% target; one does not.

Example: Add Successful Node Probabilities

The successful counts in the three-month trade are $K_3 = 2$ and $K_3 = 3$. With $p = 0.6$, their combined probability is:

$$\begin{aligned}\mathbb{P}(\rho_T > 0.05) &= \mathbb{P}(K_3 \geq 2) \\ &= \binom{3}{2} (0.6)^2 (0.4) + \binom{3}{3} (0.6)^3 \\ &= 0.432 + 0.216 = 0.648.\end{aligned}$$

The model assigns a **64.8% probability** to exceeding the target at the scheduled sale time.

The return inequality determines which counts succeed. The probability p determines how much probability those counts carry.

Computing the Cumulative Probability

The up-move count ranges from 0 to N . Record the cutoff as:

$$k_{\min} = \min\{\max\{\kappa, 0\}, N + 1\}.$$

For $\rho_{\star} > -1$, add the probabilities of the successful counts:

$$\mathbb{P}(\rho_T > \rho_{\star}) = \begin{cases} 1, & k_{\min} = 0 \quad (\text{all nodes}), \\ \sum_{k=k_{\min}}^N \binom{N}{k} p^k (1-p)^{N-k}, & 1 \leq k_{\min} \leq N, \\ 0, & k_{\min} = N + 1 \quad (\text{no nodes}). \end{cases}$$

The sum is the **binomial upper tail**. For $\rho_{\star} \leq -1$, the probability is one because every model price is positive.

What About Falling Short of the Target?

At the scheduled sale time, the scaled NPV either exceeds the target or finishes at or below it. These events are complementary:

$$\mathbb{P}(\rho_T \leq \rho_\star) = 1 - \mathbb{P}(\rho_T > \rho_\star).$$

For $\rho_\star = -0.05$, this gives the probability that the discounted fractional return finishes at or below -5% .

Increasing g_y with the lattice and target fixed cannot increase the success probability. Changing N changes both the cutoff and the count distribution.

Scheduled sale: this calculation checks the final outcome. Selling when a boundary is first reached requires the price path; see the optional first-passage example.

Example: Terminal Targets from Data

Example: Explore a terminal target probability

How likely are we to exceed our target when we fit the lattice to historical data?

- Estimate the up and down factors and up-move probability.
- Identify terminal nodes whose scaled NPV exceeds the target.
- Add their probabilities and compute the complement.

Next question: what changes if each step allows more than two price movements?

A Three-Outcome Price Model

An N-ary lattice allows m outcomes per step. Suppose $S_0 = 100$ USD/share. With $m = 3$, the price can increase by 10%, remain unchanged, or decrease by 10%.

Movement	Factor f_j	Probability p_j	Price after one step
Up 10%	1.10	0.25	110
Unchanged	1.00	0.50	100
Down 10%	0.90	0.25	90

Up then down gives $100(1.10)(0.90) = 99$ USD/share. Reversing the order gives the same price. Both have counts $(1, 0, 1)$ for up, unchanged, and down moves, so they share one node.

Computing an N-Ary Node's Price

Allow $m \geq 2$ branches with fixed price factors $f_j > 0$ and probabilities $p_j > 0$, where $\sum_{j=1}^m p_j = 1$. At level t , let x_j count branch j :

$$\mathbf{x} = (x_1, \dots, x_m), \quad x_j \in \mathbb{Z}_{\geq 0}, \quad \sum_{j=1}^m x_j = t.$$

Each occurrence of branch j multiplies the price by f_j :

$$\underbrace{S_t(\mathbf{x})}_{\text{node price}} = S_0 \prod_{j=1}^m f_j^{x_j}.$$

For our three-outcome model, $S_2(1, 0, 1) = 100(1.10)(0.90) = 99$ USD/share. The counts determine the final price; the order can change intermediate prices.

Computing an N-Ary Node's Probability

Assume independent branch choices with fixed probabilities. Let $\mathbf{X}_t$ be the random vector of branch counts after t steps.

Up-down and down-up each have probability $0.25^2 = 0.0625$. Both reach $(1, 0, 1)$, so this node has probability 0.125.

In general, add the probabilities of all paths with the same counts:

$$\mathbb{P}(\mathbf{X}_t = \mathbf{x}) = \underbrace{\frac{t!}{x_1! \cdots x_m!}}_{\text{number of orderings}} \underbrace{\prod_{j=1}^m p_j^{x_j}}_{\text{probability of one ordering}} .$$

With $m = 2$, this reduces to the binomial probability $\binom{t}{k} p^k (1-p)^{t-k}$.

Nine Paths Recombine into Six Count States

After two steps, there are $3^2 = 9$ ordered paths. We only need six count states:

Counts (up, unchanged, down)	Paths	Price (USD/share)	Probability
(2, 0, 0)	1	121	0.0625
(1, 1, 0)	2	110	0.2500
(1, 0, 1)	2	99	0.1250
(0, 2, 0)	1	100	0.2500
(0, 1, 1)	2	90	0.2500
(0, 0, 2)	1	81	0.0625

The six probabilities sum to one and give the two-step price distribution.

Example: A Target in the Three-Outcome Model

Use the preceding table with $S_0 = 100$ USD/share, $N = 2$ monthly steps, $T = 1/6$ year, and $g_y = 0.04 \text{ year}^{-1}$.

A target scaled NPV of 5% requires:

$$S_T > S_0(1 + \rho_\star)e^{g_y T} = 100(1.05)e^{0.04/6} \approx 105.70 \text{ USD/share.}$$

Only $(2, 0, 0)$ and $(1, 1, 0)$, priced at 121 and 110 USD/share, qualify. Add their probabilities:

$$\mathbb{P}(\rho_T > 0.05) = 0.0625 + 0.2500 = 0.3125.$$

The model assigns a **31.25% probability** to exceeding the 5% target at the scheduled sale time, two months after purchase.

How Many States Do We Need?

A count state satisfies $x_1 + \cdots + x_m = t$. Represent the t moves by stars and separate the m branch groups using $m - 1$ bars:

$$\star \mid \mid \star \quad \longleftrightarrow \quad (1, 0, 1).$$

Choose the bar positions among $t + m - 1$ available positions:

$$L_t = \binom{t + m - 1}{m - 1}, \quad L_2 = \binom{4}{2} = 6 \quad \text{when } m = 3.$$

This counts branch-count states, not necessarily distinct prices. If $f_1 f_3 = f_2^2$, for example, $(1, 0, 1)$ and $(0, 2, 0)$ share a price. We add their probabilities when computing the probability of that price.

Example: N-Ary Lattices from Data

Example: [Explore N-ary lattice models](#)

How do we build a lattice with several branches from historical data?

- Estimate branch factors and probabilities from historical growth rates.
- Build the lattice and calculate node prices and multinomial probabilities.
- Check that probabilities sum to one and plot the resulting distributions.

Toward continuous time: lattices can approach continuous-time price models when price changes are scaled as the time step shrinks.

Optional Advanced Material

Advanced: [First-Passage Exit Rules](#)

What changes if we check the price after every step? We track positions until a take-profit or stop-loss boundary is first reached, then compare with checking only the final price.

Advanced: [Execution-Aware Probability of Profit](#)

How do trading costs change the target probability? We include the bid–ask spread, fees, and unfavorable execution prices (slippage), then compare the result with the frictionless calculation.

Optional; neither notebook is a prerequisite for L4b.

Summary

We used net present value and lattice probabilities to evaluate a long stock position.

Key takeaways

- **NPV-based trading framework:** We scaled the NPV by the initial investment to compare discounted sale proceeds with the purchase cost.
- **Terminal target probabilities:** We found the minimum up-move count needed to exceed a target scaled NPV, then added the successful node probabilities. The complement gives the probability of finishing at or below the target.
- **N-ary lattice models:** We used branch counts to calculate prices, multinomial probabilities, and state counts for m outcomes per step. More branches increase the computational cost.

Next time: continuous-time stochastic price models.

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